

Soft Computing — Unit 1: Introduction

Syllabus (8 Hrs): Introduction to soft computing; introduction to fuzzy sets and fuzzy logic systems; introduction to biological and artificial neural networks; introduction to Genetic Algorithm.

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1. Introduction to Soft Computing

1.1 Theory

Soft Computing (SC) is a collection of computational techniques in computer science that — *unlike conventional (hard) computing* — is **tolerant of imprecision, uncertainty, partial truth, and approximation**. The guiding principle, given by **Prof. Lotfi A. Zadeh (1994)**, is:

"The role model for soft computing is the human mind."

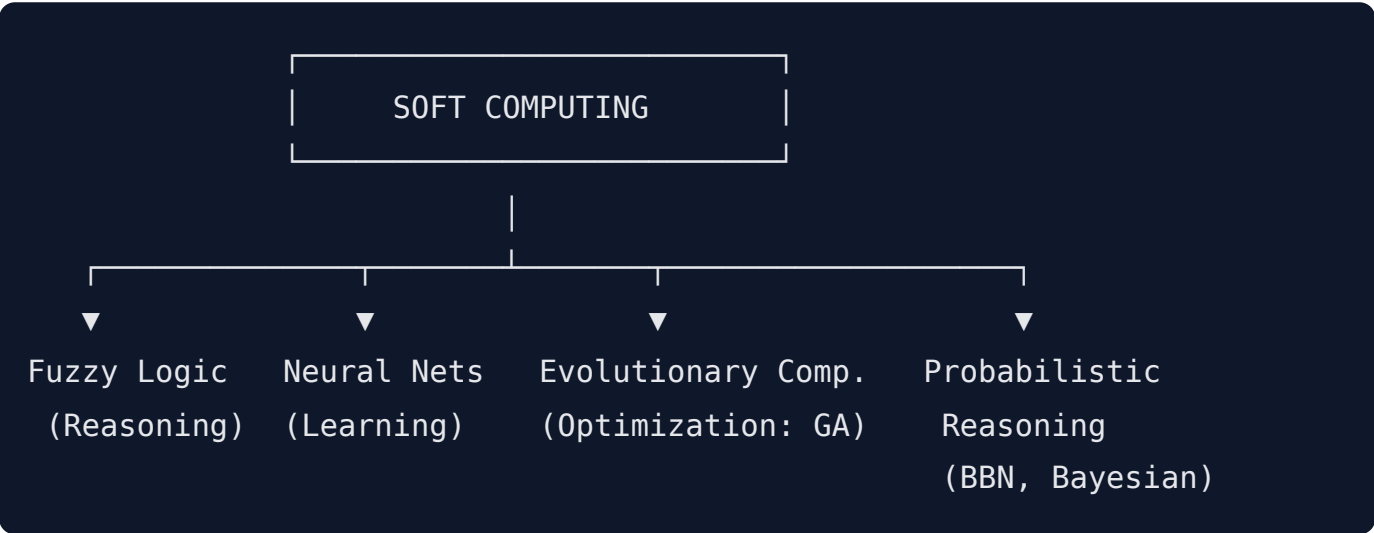
The aim is to **exploit tolerance for imprecision** to achieve tractability, robustness, low solution-cost, and a better rapport with reality.

Characteristics of Soft Computing

Property	Meaning
Approximation	Solutions are <i>good-enough</i> , not necessarily exact.

Uncertainty handling	Works with vague/noisy/partial data.
Learning ability	Improves from examples (data-driven).
Bio-inspired	Mimics biological & evolutionary processes.
Parallelism	Naturally parallel architectures (e.g., neural nets).
Robustness	Performs gracefully under noise & faults.

Components (Pillars) of Soft Computing



1. **Fuzzy Logic (FL)** — Approximate reasoning with linguistic variables.
 2. **Artificial Neural Networks (ANN)** — Learning from data; pattern recognition.
 3. **Evolutionary Computation (EC)** — Population-based optimization (GA, GP, ES).
 4. **Probabilistic Reasoning (PR)** — Bayesian nets, belief networks, chaos theory.
- These methods are **complementary** rather than competing — they are often combined into **Hybrid Soft Computing Systems** (e.g., Neuro-Fuzzy, Genetic-Fuzzy, Neuro-Genetic).

1.2 Hard Computing vs Soft Computing

Feature	Hard Computing	Soft Computing
Basis	Binary / Boolean logic	Multi-valued / Fuzzy logic
Precision	Exact answers required	Approximate answers acceptable

Computation	Sequential	Parallel
Data	Precise data	Imprecise / noisy data
Model	Mathematical (deterministic)	Empirical / bio-inspired
Cost	High	Low
Examples	Numerical methods, sorting, search	ANN, Fuzzy, GA



1.3 Mathematical Foundation

A **soft-computing system** can be viewed as a mapping:

$$f : X \subseteq \mathbb{R}^n \longrightarrow Y \subseteq \mathbb{R}^m$$

approximated by a parameterized family $\hat{f}(\mathbf{x}; \boldsymbol{\theta})$ where the parameter vector $\boldsymbol{\theta}$ is **learned** or **evolved** to minimize an error functional:

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} E(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{f}(\mathbf{x}_i; \boldsymbol{\theta}))$$

where $L(\cdot, \cdot)$ is a loss function.



1.4 Applications

- Handwriting / speech / face recognition
- Medical diagnosis (e.g., fuzzy cancer-risk classifier)
- Weather forecasting
- Autonomous driving & robotics
- Stock-market prediction
- Smart appliances (washing machines, ACs, rice cookers)
- Image compression & enhancement
- Power-system load forecasting

2. Fuzzy Sets and Fuzzy Logic Systems

2.1 Need for Fuzzy Logic

Real-world concepts ("*tall man*", "*warm water*", "*fast car*") cannot be modelled with **crisp** (yes/no) sets. **Lotfi Zadeh (1965)** proposed **Fuzzy Sets** to capture *degrees of membership*.

Crisp Set vs Fuzzy Set

	Crisp Set A	Fuzzy Set $\tilde{A}$
Membership $\mu_A(x)$	$\{0, 1\}$	$[0, 1]$
Boundary	Sharp	Gradual
Notation	$A = \{x : P(x)\}$	$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) : x \in X\}$

2.2 Formal Definitions

Fuzzy Set

A fuzzy set $\tilde{A}$ over a universe of discourse X is defined by its **membership function**:

$$\mu_{\tilde{A}} : X \longrightarrow [0, 1]$$

Discrete representation:

$$\tilde{A} = \sum_{x_i \in X} \frac{\mu_{\tilde{A}}(x_i)}{x_i} \quad (\text{Zadeh's notation; '+' and '/' are NOT arithmetic})$$

Continuous representation:

$$\tilde{A} = \int_X \frac{\mu_{\tilde{A}}(x)}{x}$$

Support, Core, Height, α -Cut

Term	Definition
Support	$\text{supp}(\tilde{A}) = \{x \in X : \mu_{\tilde{A}}(x) > 0\}$
Core	$\text{core}(\tilde{A}) = \{x \in X : \mu_{\tilde{A}}(x) = 1\}$
Height	$h(\tilde{A}) = \sup_{x \in X} \mu_{\tilde{A}}(x)$
Normal set	$h(\tilde{A}) = 1$

α-cut	$A_\alpha = \{x : \mu_{\tilde{A}}(x) \geq \alpha\}$
Crossover point	x such that $\mu_{\tilde{A}}(x) = 0.5$



2.3 Common Membership Functions

Triangular

$$\mu(x; a, b, c) = \max \left(\min \left(\frac{x - a}{b - a}, \frac{c - x}{c - b} \right), 0 \right)$$

Trapezoidal

$$\mu(x; a, b, c, d) = \max \left(\min \left(\frac{x - a}{b - a}, 1, \frac{d - x}{d - c} \right), 0 \right)$$

Gaussian

$$\mu(x; c, \sigma) = \exp \left(-\frac{(x - c)^2}{2\sigma^2} \right)$$

Sigmoidal

$$\mu(x; a, c) = \frac{1}{1 + e^{-a(x - c)}}$$

Bell-shaped (Generalized)

$$\mu(x; a, b, c) = \frac{1}{1 + \left| \frac{x - c}{a} \right|^{2b}}$$



2.4 Operations on Fuzzy Sets

Let $\tilde{A}, \tilde{B}$ be fuzzy sets over X .

Operation	Definition (Zadeh's standard)
Union	$\mu_{\tilde{A} \cup \tilde{B}}(x) = \max(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))$
Intersection	$\mu_{\tilde{A} \cap \tilde{B}}(x) = \min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))$
Complement	$\mu_{\tilde{A}^c}(x) = 1 - \mu_{\tilde{A}}(x)$

Equality	$\mu_{\tilde{A}}(x) = \mu_{\tilde{B}}(x) \forall x$
Containment	$\tilde{A} \subseteq \tilde{B} \iff \mu_{\tilde{A}}(x) \leq \mu_{\tilde{B}}(x) \forall x$
Product	$\mu_{\tilde{A} \cdot \tilde{B}}(x) = \mu_{\tilde{A}}(x) \cdot \mu_{\tilde{B}}(x)$
Algebraic sum	$\mu_{\tilde{A} + \tilde{B}}(x) = \mu_{\tilde{A}}(x) + \mu_{\tilde{B}}(x) - \mu_{\tilde{A}}(x)\mu_{\tilde{B}}(x)$
Bounded sum	$\mu(x) = \min(1, \mu_{\tilde{A}}(x) + \mu_{\tilde{B}}(x))$
Bounded difference	$\mu(x) = \max(0, \mu_{\tilde{A}}(x) - \mu_{\tilde{B}}(x))$

t-norms (generalized intersection): min, product, Łukasiewicz $\max(0, a + b - 1)$.

t-conorms / s-norms (generalized union): max, algebraic sum, $\min(1, a + b)$.

2.5 Important Properties / Theorems

For fuzzy sets, **most** of the classical set-theoretic laws hold (with max / min semantics):

Law	Statement
Commutativity	$\tilde{A} \cup \tilde{B} = \tilde{B} \cup \tilde{A}$
Associativity	$(\tilde{A} \cup \tilde{B}) \cup \tilde{C} = \tilde{A} \cup (\tilde{B} \cup \tilde{C})$
Distributivity	$\tilde{A} \cap (\tilde{B} \cup \tilde{C}) = (\tilde{A} \cap \tilde{B}) \cup (\tilde{A} \cap \tilde{C})$
Idempotency	$\tilde{A} \cup \tilde{A} = \tilde{A}$
Identity	$\tilde{A} \cup \emptyset = \tilde{A}, \tilde{A} \cap X = \tilde{A}$
De Morgan's	$\overline{\tilde{A} \cup \tilde{B}} = \tilde{A} \cap \tilde{B}$
Involution	$\overline{\tilde{\tilde{A}}} = \tilde{A}$


Important exception: The **Law of Excluded Middle** and **Law of Contradiction** **DO NOT** hold for fuzzy sets:

$\tilde{A} \cup \tilde{\tilde{A}} \neq X, \quad \tilde{A} \cap \tilde{\tilde{A}} \neq \emptyset$

Decomposition (Representation) Theorem

Any fuzzy set $\tilde{A}$ can be represented as the union of its α -cuts:

$$\tilde{A} = \bigcup_{\alpha \in (0,1]} \alpha \cdot A_\alpha \quad \text{where} \quad \mu_{\alpha A_\alpha}(x) = \begin{cases} \alpha & x \in A_\alpha \\ 0 & \text{otherwise} \end{cases}$$

 Extension Principle (Zadeh)

If $f : X \rightarrow Y$ and $\tilde{A}$ is a fuzzy set on X , then the fuzzy set $\tilde{B} = f(\tilde{A})$ on Y is:

$$\mu_{\tilde{B}}(y) = \sup_{x: y=f(x)} \mu_{\tilde{A}}(x)$$

 2.6 Solved Problems

 Problem 1 — Basic Operations

Given $X = \{1, 2, 3, 4, 5\}$ and

$$\begin{aligned} \tilde{A} &= \frac{0.2}{1} + \frac{0.5}{2} + \frac{0.8}{3} + \frac{1.0}{4} + \frac{0.3}{5} \\ \tilde{B} &= \frac{0.6}{1} + \frac{0.4}{2} + \frac{0.7}{3} + \frac{0.5}{4} + \frac{0.9}{5} \end{aligned}$$

Find $\tilde{A} \cup \tilde{B}, \tilde{A} \cap \tilde{B}, \bar{\tilde{A}}$.

Solution:

x	$\mu_{\tilde{A}}$	$\mu_{\tilde{B}}$	max (Union)	min (Intersection)	$1 - \mu_{\tilde{A}}$
1	0.2	0.6	0.6	0.2	0.8
2	0.5	0.4	0.5	0.4	0.5
3	0.8	0.7	0.8	0.7	0.2
4	1.0	0.5	1.0	0.5	0.0
5	0.3	0.9	0.9	0.3	0.7

$$\begin{aligned} \tilde{A} \cup \tilde{B} &= \frac{0.6}{1} + \frac{0.5}{2} + \frac{0.8}{3} + \frac{1.0}{4} + \frac{0.9}{5} \\ \tilde{A} \cap \tilde{B} &= \frac{0.2}{1} + \frac{0.4}{2} + \frac{0.7}{3} + \frac{0.5}{4} + \frac{0.3}{5} \\ \bar{\tilde{A}} &= \frac{0.8}{1} + \frac{0.5}{2} + \frac{0.2}{3} + \frac{0.0}{4} + \frac{0.7}{5} \end{aligned}$$



Problem 2 — α -Cut

Using $\tilde{A}$ above, find $A_{0.5}$ and $A_{0.8}$.

Solution: Pick elements with $\mu \geq \alpha$.

- $A_{0.5} = \{2, 3, 4\}$ (since $\mu = 0.5, 0.8, 1.0$).
 - $A_{0.8} = \{3, 4\}$.
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Problem 3 — Triangular Membership

Define "warm" temperature with triangular MF (20, 25, 30). Find $\mu(22)$ and $\mu(27)$.

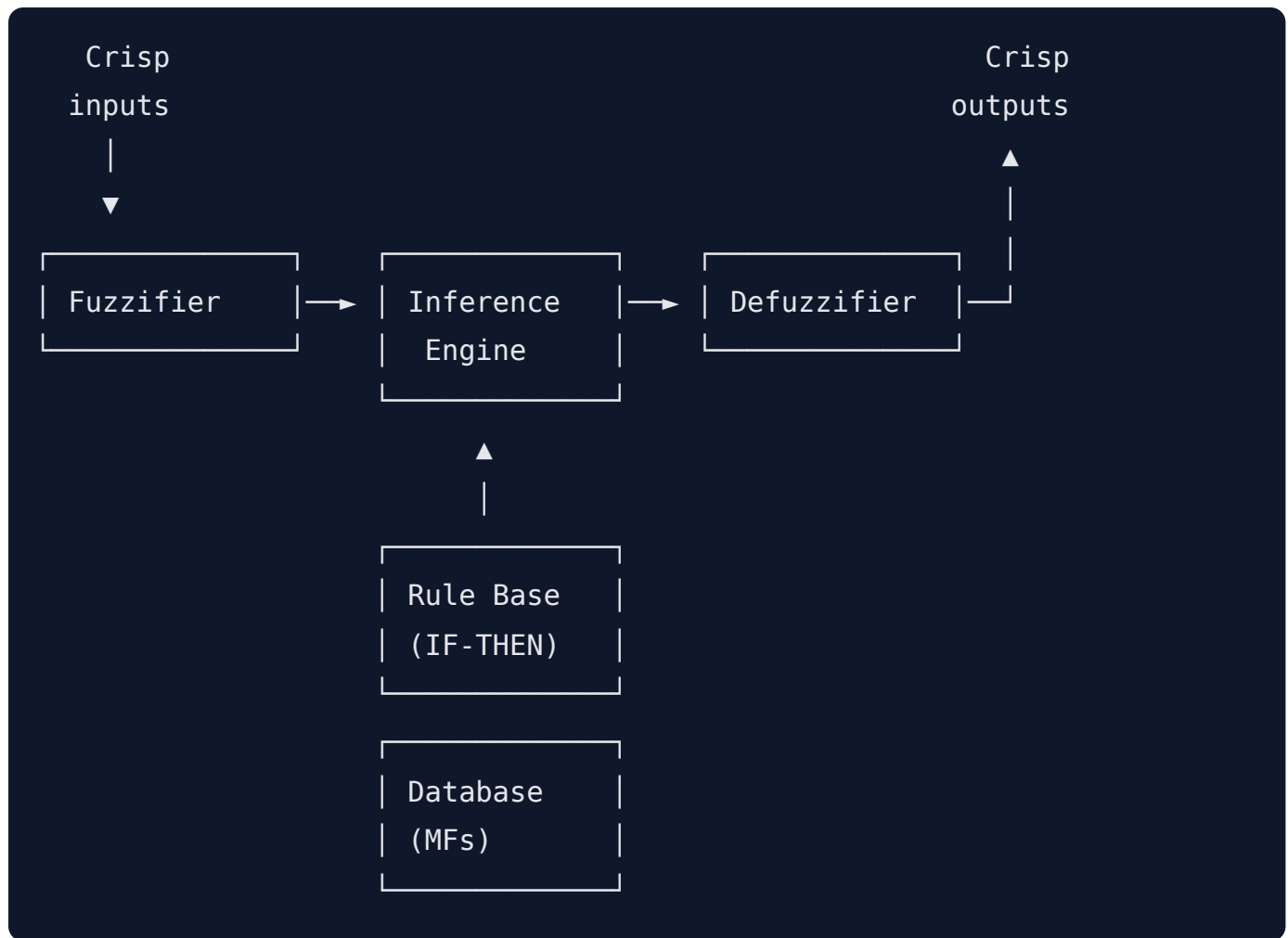
$$\mu(22) = \frac{22 - 20}{25 - 20} = \frac{2}{5} = 0.4$$

$$\mu(27) = \frac{30 - 27}{30 - 25} = \frac{3}{5} = 0.6$$



2.7 Fuzzy Logic Systems (FLS)

A **Fuzzy Inference System (FIS)** maps crisp inputs to crisp outputs via fuzzy rules.



Stages

1. **Fuzzification** — Crisp $x \rightarrow$ fuzzy membership values.
2. **Rule Base** — IF-THEN rules from expert knowledge.
Example: IF temperature is HIGH AND humidity is HIGH THEN fan-speed is FAST.
3. **Inference Engine** — Combines rules using **Mamdani** or **Sugeno** methods.
4. **Defuzzification** — Fuzzy output $\rightarrow$ crisp value.

Defuzzification Methods

Method	Formula
Centroid (COG)	$x^* = \frac{\int x\mu(x)dx}{\int \mu(x)dx}$
Bisector	$x^* \text{ such that } \int_a^{x^*} \mu(x)dx = \int_{x^*}^b \mu(x)dx$
Mean of Maxima (MOM)	Average x where $\mu(x)$ peaks
Smallest of Max (SOM)	Smallest x where μ is max

Largest of Max (LOM)	Largest x where μ is max
Weighted Average	$x^* = \frac{\sum w_i x_i}{\sum w_i}$

Problem 4 — Centroid Defuzzification

Fuzzy output:

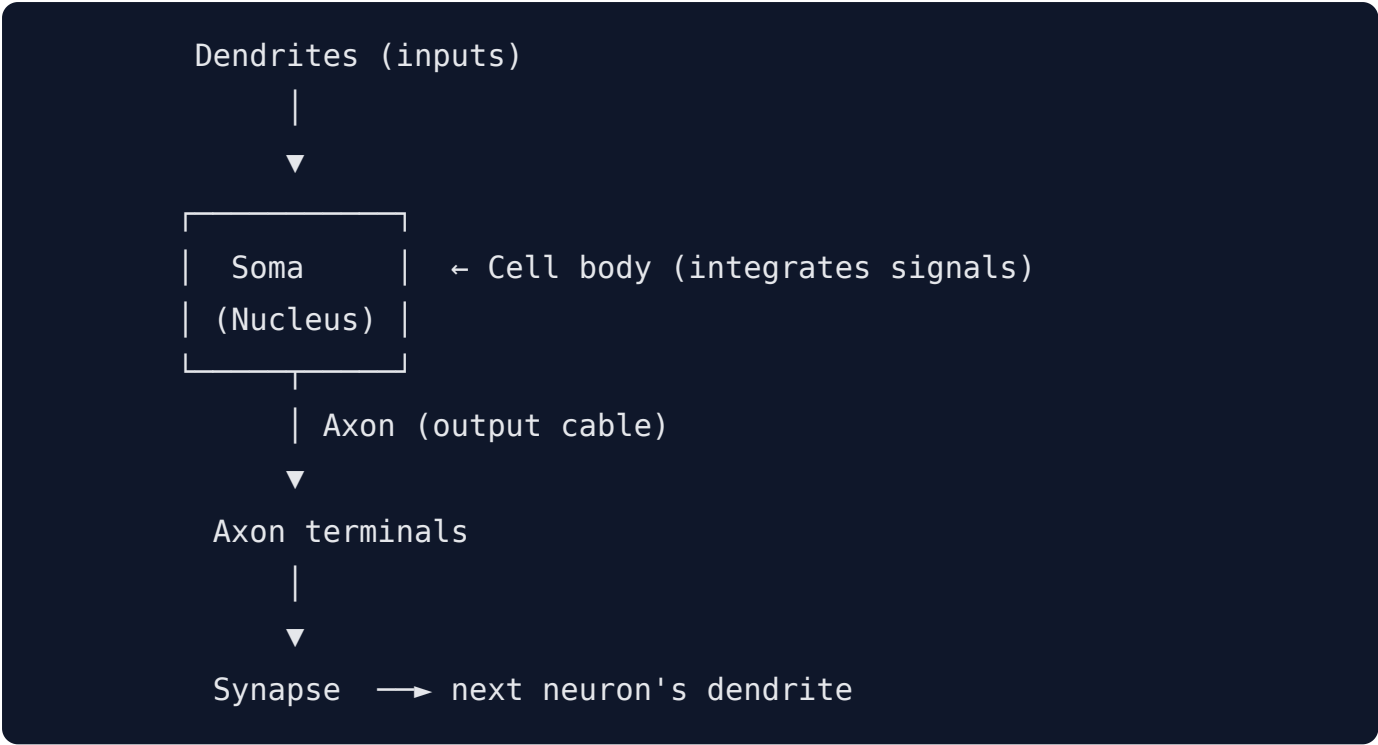
x	1	2	3	4	5
$\mu(x)$	0.2	0.5	0.8	0.6	0.3

$$\begin{aligned}
 x^* &= \frac{\sum x_i \mu(x_i)}{\sum \mu(x_i)} = \frac{1(0.2) + 2(0.5) + 3(0.8) + 4(0.6) + 5(0.3)}{0.2 + 0.5 + 0.8 + 0.6 + 0.3} \\
 &= \frac{0.2 + 1.0 + 2.4 + 2.4 + 1.5}{2.4} = \frac{7.5}{2.4} \approx \mathbf{3.125}
 \end{aligned}$$

3. Biological and Artificial Neural Networks

3.1 Biological Neuron

The human brain contains $\approx 10^{11}$ **neurons** with $\approx 10^{14}$ **synaptic connections**.



Component	Function
Dendrites	Receive signals from other neurons
Soma (cell body)	Sums incoming signals; if threshold crossed → fires
Axon	Carries output signal away
Synapse	Junction; modulates signal strength (weight)
Myelin sheath	Insulator, speeds up signal propagation

Signal: electrochemical impulse (**action potential**) — "*all-or-none*" law.

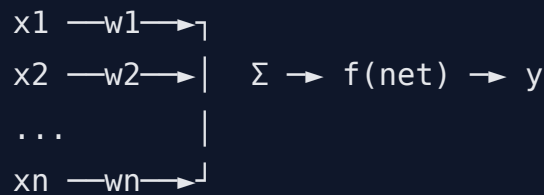
3.2 Biological vs Artificial Neuron

Biological Neuron	Artificial Neuron
Dendrite	Input x_i
Synapse	Weight w_i
Soma	Summation Σ + activation f
Axon	Output y

~ms response	ns – μ s response
Highly parallel	Mostly simulated parallelism
$\sim 10^{11}$ neurons	$10^2 - 10^9$ (in DNNs)
Adapts (plasticity)	Adapts via learning algorithm

3.3 McCulloch–Pitts Neuron (1943)

The earliest mathematical neuron model:



Net input:

$$\text{net} = \sum_{i=1}^n w_i x_i - \theta$$

Output (binary step):

$$y = f(\text{net}) = \begin{cases} 1, & \text{net} \geq 0 \\ 0, & \text{net} < 0 \end{cases}$$

where θ is the **threshold** (bias).

3.4 Activation Functions

Name	Formula	Range
Step / Heaviside	$f(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$	$\{0, 1\}$
Signum	$f(x) = \text{sgn}(x)$	$\{-1, 0, 1\}$
Linear	$f(x) = x$	$\mathbb{R}$

Sigmoid (logistic)	$f(x) = \frac{1}{1 + e^{-x}}$	$(0, 1)$
Tanh	$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$(-1, 1)$
ReLU	$f(x) = \max(0, x)$	$[0, \infty)$
Leaky ReLU	$f(x) = \max(0.01x, x)$	$\mathbb{R}$
Softmax	$f_i(\mathbf{x}) = \frac{e^{x_i}}{\sum_j e^{x_j}}$	$(0, 1)$, sums to 1

Sigmoid derivative (useful for BP): $f'(x) = f(x)(1 - f(x))$.

3.5 Architectures of ANN

1. **Single-Layer Feed-Forward** — Perceptron.
2. **Multi-Layer Feed-Forward (MLP)** — Input, hidden, output layers; trained by back-propagation.
3. **Recurrent (RNN)** — Contains feedback loops; handles sequential data (Hopfield, LSTM).
4. **Self-Organizing Map (SOM, Kohonen)** — Unsupervised; topology-preserving.
5. **Radial Basis Function (RBF) Network** — Local activation, fast learning.
6. **Convolutional NN (CNN)** — For images.

3.6 Learning Paradigms

Type	Description	Examples
Supervised	Input + target given	Perceptron, BP, RBF
Unsupervised	Only input, find structure	SOM, Hebbian, ART
Reinforcement	Reward / penalty feedback	Q-learning, TD

Learning Rules

Rule	Update equation
------	-----------------

Hebbian	$\Delta w_{ij} = \eta x_i y_j$
Perceptron	$\Delta w_i = \eta(t - y)x_i$
Delta (Widrow–Hoff)	$\Delta w_i = \eta(t - y)f'(\text{net})x_i$
Competitive	$\Delta w_{ij} = \eta(x_i - w_{ij})$ if neuron j wins
Correlation	$\Delta w_{ij} = \eta x_i t_j$
Outstar (Grossberg)	$\Delta w_{ij} = \eta(t_j - w_{ij})$

Here η is the **learning rate**, t is target, y is actual output.

3.7 Solved Problems

Problem 5 — McCulloch–Pitts AND Gate

Design an M-P neuron that computes the logical AND of two binary inputs x_1, x_2 .

Solution: Take $w_1 = w_2 = 1, \theta = 2$.

x_1	x_2	net = $x_1 + x_2 - 2$	y
0	0	-2	0
0	1	-1	0
1	0	-1	0
1	1	0	1

Output matches AND truth-table.

Problem 6 — Perceptron Learning (1 step)

Initial weights $w_1 = 0.2, w_2 = -0.1, b = 0.0$. Input $(x_1, x_2) = (1, 1)$, target $t = 1$, learning rate $\eta = 0.5$, activation = step.

Step 1: net input

$$\text{net} = (1)(0.2) + (1)(-0.1) + 0 = 0.1 \geq 0 \Rightarrow y = 1$$

Since $y = t$, weights unchanged.

Now feed $(1, 0)$, target $t = 0$:

$$\text{net} = 0.2 + 0 + 0 = 0.2 \geq 0 \Rightarrow y = 1$$

Error $e = t - y = -1$. Update:

$$w_1 \leftarrow 0.2 + (0.5)(-1)(1) = -0.3$$

$$w_2 \leftarrow -0.1 + (0.5)(-1)(0) = -0.1$$

$$b \leftarrow 0 + (0.5)(-1) = -0.5$$

New weights: $(w_1, w_2, b) = (-0.3, -0.1, -0.5)$.

3.8 Important Theorems

Perceptron Convergence Theorem (Rosenblatt, 1962)

If the training set is **linearly separable**, the perceptron learning algorithm converges in a **finite** number of steps to a weight vector that classifies all examples correctly.

Universal Approximation Theorem (Cybenko 1989; Hornik 1991)

A feed-forward network with a **single hidden layer** containing a *finite* number of neurons and a **non-constant, bounded, continuous** activation function (e.g., sigmoid) can approximate any continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ on a compact set to **arbitrary precision**.

XOR Limitation (Minsky & Papert, 1969)

A **single-layer perceptron cannot** solve the XOR problem because XOR is **not linearly separable**. At least one hidden layer is required.

4. Genetic Algorithms

4.1 Inspiration

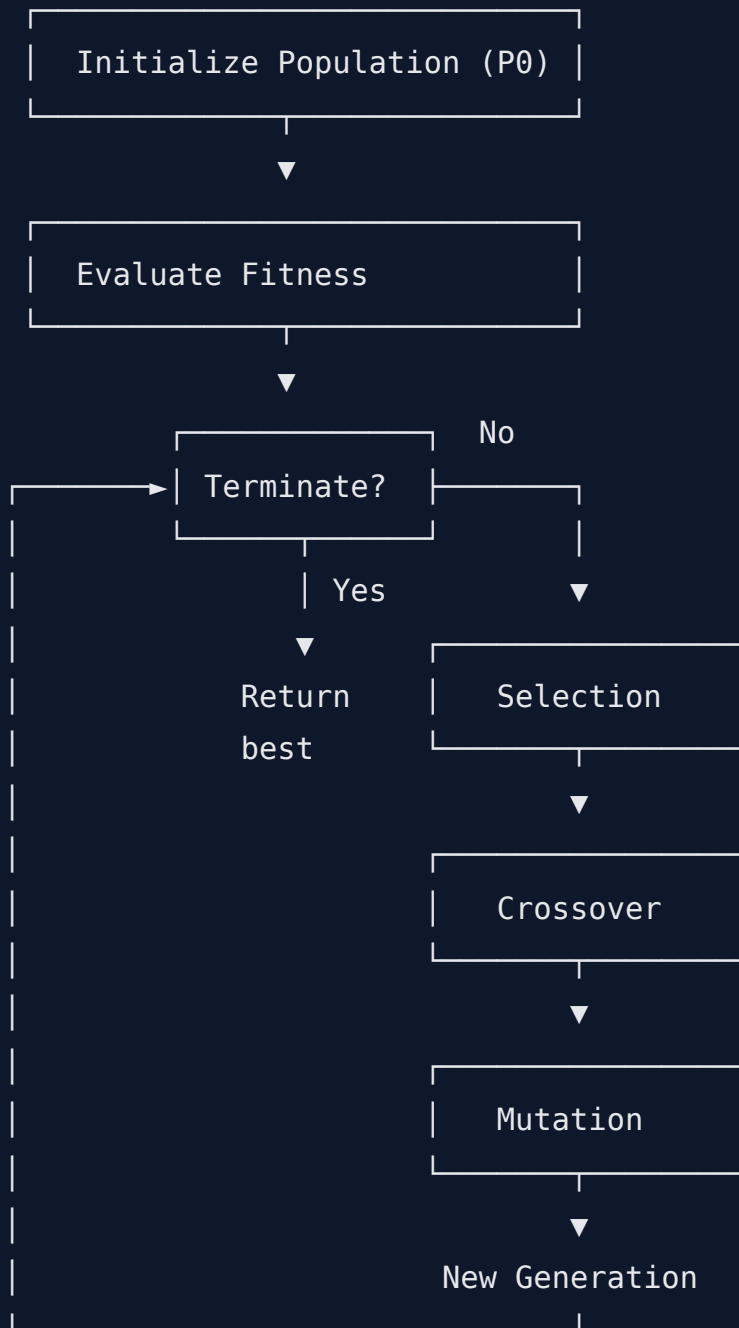
Proposed by John Holland (1975) — a **population-based metaheuristic** inspired by **Darwinian natural selection** and **Mendelian genetics**.

Key idea: Encode candidate solutions as *chromosomes*. Evolve them through *selection*, *crossover*, and *mutation* across generations; better solutions (higher *fitness*) survive and reproduce.

4.2 Terminology

GA Term	Biology	Meaning
Chromosome	DNA strand	Encoded candidate solution (string of genes)
Gene	DNA segment	One variable / bit
Allele	Variant of gene	Possible value of a gene
Locus	Position on chromosome	Index in string
Genotype	Encoded form	Bit string 10110
Phenotype	Decoded form	Real-world value (e.g., $x = 22$)
Population	Group	Set of chromosomes
Fitness	Survival ability	Objective-function value
Generation	Lifecycle	One iteration

4.3 GA Workflow



Algorithm (Pseudo-code)

```

Initialize population  $P(0)$  randomly
Evaluate fitness of each individual in  $P(0)$ 
 $t \leftarrow 0$ 
while (termination condition not met) do
    Select parents from  $P(t)$  using a selection scheme
    Apply crossover with probability  $p_c \rightarrow$  offspring
    Apply mutation with probability  $p_m \rightarrow$  mutated offspring
    Evaluate fitness of offspring
    Form  $P(t+1)$  by replacement (elitism / generational)
     $t \leftarrow t + 1$ 
return best individual in  $P(t)$ 

```



4.4 GA Operators

Selection

Scheme	Idea
Roulette Wheel	$p_i = \frac{f_i}{\sum_j f_j}$
Tournament	Pick k random; choose fittest
Rank-based	Sort by fitness; assign probabilities by rank
Stochastic Universal Sampling	N equally-spaced pointers on wheel
Elitism	Carry best e individuals unchanged

Crossover (Recombination)

Type	Description
Single-point	Pick locus k ; swap suffixes after k
Two-point	Pick two loci; swap middle segment
Uniform	For each gene, swap with probability 0.5
Arithmetic	$c = \alpha p_1 + (1 - \alpha)p_2$ for real-valued

Mutation

Type	Description
Bit-flip	Flip a bit with probability p_m
Swap	Swap two genes
Inversion	Reverse a sub-string
Gaussian	Add $\mathcal{N}(0, \sigma^2)$ noise (real-valued)

Probabilities — typical values

- Crossover: $p_c \in [0.6, 0.95]$
- Mutation: $p_m \in [0.001, 0.05]$



4.5 Schema Theorem (Holland, 1975)

A **schema** H is a template (e.g., $1*0*1$) representing a similarity subset of chromosomes.

Define:

- $m(H, t)$ = number of strings matching H at generation t
- $f(H)$ = average fitness of strings matching H
- $\bar{f}$ = average fitness of population
- $o(H)$ = **order** = # of fixed (non- $*$) positions
- $\delta(H)$ = **defining length** = distance between first and last fixed position
- L = chromosome length, p_c = crossover prob, p_m = mutation prob

$$m(H, t+1) \geq m(H, t) \cdot \frac{f(H)}{\bar{f}} \cdot \left[1 - p_c \frac{\delta(H)}{L-1}\right] \cdot [1 - p_m]^{o(H)}$$

Building Block Hypothesis (corollary): GAs work by combining **short, low-order, high-fitness** schemata (*building blocks*) into better solutions.



4.6 Solved Problem — Maximize $f(x) = x^2$ on

$\{0, \dots, 31\}$

Encoding: 5-bit binary string $\Rightarrow$ chromosome length $L = 5$.

Initial population (random):

Chromosome	Decoded x	$f(x) = x^2$
01101	13	169
11000	24	576
01000	8	64
10011	19	361
Total		1170
Avg		292.5

Roulette-wheel selection probabilities:

String	f_i	$p_i = f_i / \sum f_j$	Expected count $f_i / \bar{f}$
01101	169	0.144	0.58
11000	576	0.492	1.97
01000	64	0.055	0.22
10011	361	0.309	1.23

Assume mating pool after selection: {01101, 11000, 11000, 10011}.

Single-point crossover at point 4 between 01101 and 11000:

```
Parent 1: 0110 | 1      Child 1: 0110 | 0  → 01100 = 12
Parent 2: 1100 | 0      Child 2: 1100 | 1  → 11001 = 25
```

Crossover at point 2 between 11000 and 10011:

```
Parent 1: 11 | 000      Child 1: 11 | 011 → 11011 = 27
Parent 2: 10 | 011      Child 2: 10 | 000 → 10000 = 16
```

Mutation: flip bit 3 of 11011 $\rightarrow$ 11111 = 31, $f = 961$. 🎉

New population:

Chromosome	x	$f(x)$
01100	12	144
11001	25	625
11111	31	961
10000	16	256
Total		1986
Avg		496.5

Average fitness rose from **292.5** $\rightarrow$ **496.5**; best individual reached the **optimum** $x = 31$. ✅

📖 4.7 Advantages & Limitations of GA

✅ Advantages

- Works with any objective (no derivatives needed).
- Excellent for **multi-modal**, discontinuous, non-differentiable spaces.
- Naturally **parallel**.
- Provides multiple alternative solutions.

⚠️ Limitations

- No guarantee of global optimum.
- Many hyper-parameters (p_c , p_m , pop. size).
- Slow convergence on simple problems.
- Premature convergence (loss of diversity).

5. Quick Revision Cheat-Sheet

📝 Soft Computing

- **Father:** Lotfi Zadeh.
- **4 pillars:** Fuzzy Logic, ANN, Evolutionary Computation, Probabilistic Reasoning.
- **Tolerant** of imprecision; **hard** computing is exact.



Fuzzy Sets

- Membership $\mu \in [0, 1]$.
- $\cup = \max, \cap = \min, \bar{} = 1 - \mu$.
- α -cut = $\{x : \mu(x) \geq \alpha\}$.
- **Excluded middle FAILS** in fuzzy logic.
- FIS: Fuzzify $\rightarrow$ Inference $\rightarrow$ Defuzzify (Mamdani / Sugeno).
- Centroid: $x^* = \frac{\sum x\mu(x)}{\sum \mu(x)}$.



Neural Networks

- Biological: dendrite-soma-axon-synapse.
- Artificial: $\text{net} = \sum w_i x_i - \theta, y = f(\text{net})$.
- **Activation:** step, sigmoid, tanh, ReLU.
- **Learning:** Hebbian, Perceptron, Delta.
- **Perceptron convergence** — linearly separable data only.
- **Universal approximation** — one hidden layer suffices.
- **XOR** $\Rightarrow$ need ≥ 1 hidden layer.



Genetic Algorithm

- Father: John Holland (1975).
 - Cycle: **Init** $\rightarrow$ **Eval** $\rightarrow$ **Select** $\rightarrow$ **Crossover** $\rightarrow$ **Mutate** $\rightarrow$ **Repeat**.
 - Operators: roulette/tournament selection, 1-pt/uniform crossover, bit-flip mutation.
 - $p_c \approx 0.7, p_m \approx 0.01$.
 - **Schema theorem:** short, low-order, above-average schemata survive exponentially.
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End of Unit 1 — Introduction (Soft Computing).

Next: Unit 2 will cover Fuzzy Set Theory & Fuzzy Logic in depth (share when ready).